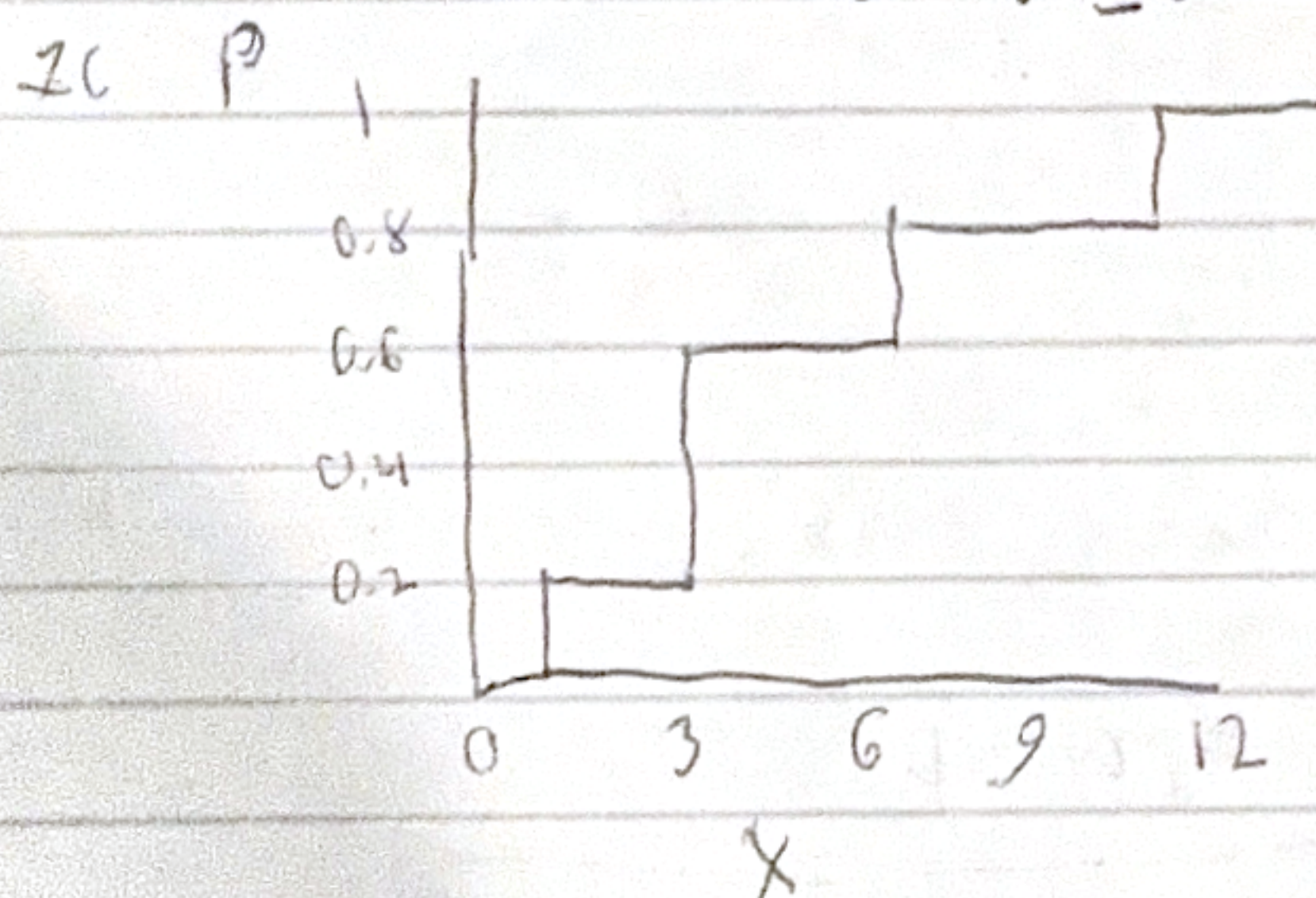


~~is red~~ ~~is blue~~ ~~is green~~

1a)	id	is_red	is_blue	is_green	is_gray
	1	1	0	0	0
	2	0	1	0	0
	3	0	1	0	0
	4	1	0	0	0
	5	0	0	1	0
	6	0	0	0	1

1b) $N_{red} = 1/3$ $N_{blue} = 1/3$ $N_{green} = 1/6$ $N_{gray} = 1/6$



2a) $Y = aX + b$ $a > 0$ $F_Y(y) = P(Y \leq y)$
 $F_Y(y) = P(aX + b \leq y)$ $F_Y(y) = P(X \leq \frac{y-b}{a})$

$F_Y(y) = F_X(\frac{y-b}{a})$ $f_Y(y) = \frac{d}{dy} F_X(\frac{y-b}{a})$
 $f_Y(y) = \frac{1}{a} f_X(\frac{y-b}{a})$

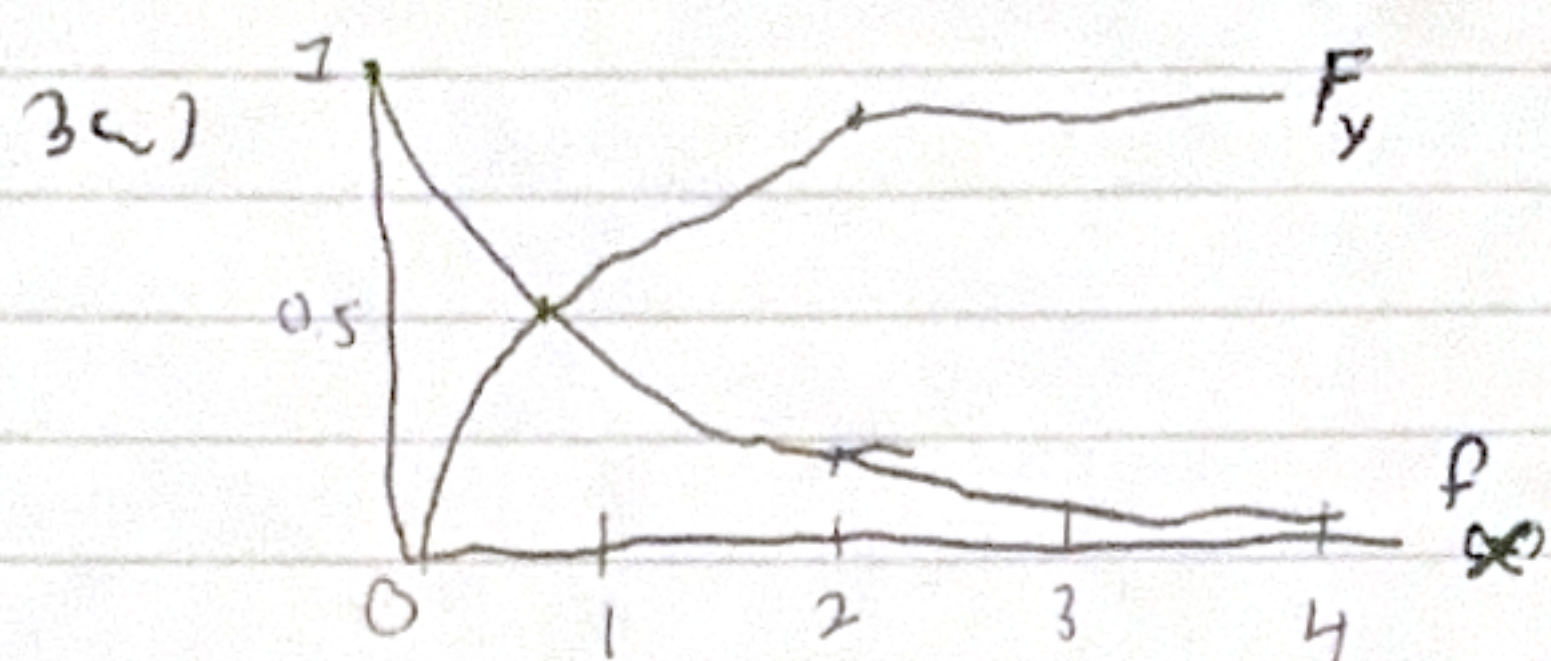
$\frac{26}{3} = \frac{78}{9}$

2b) $8/3 = N$ 2c) $\frac{1+8+16}{3} = \left(\frac{26}{3}\right)$

$\frac{8}{3} \times \frac{8}{3} = \frac{64}{9}$

$\frac{64}{9} \neq \frac{78}{9}$

2d) No



$$CDF = 1 - e^{-x}$$

$$\frac{d}{dx}(1 - e^{-x}) = 0 + e^{-x} = e^{-x}$$

$$PDF = e^{-x}$$

36) $P(X \leq 3) = 1 - P(X > 3)$

$$F_X(3) = 1 - e^{-3}$$

$$1 - F_X(3) = 1 - (1 - e^{-3}) = e^{-3} = \frac{1}{e^3}$$

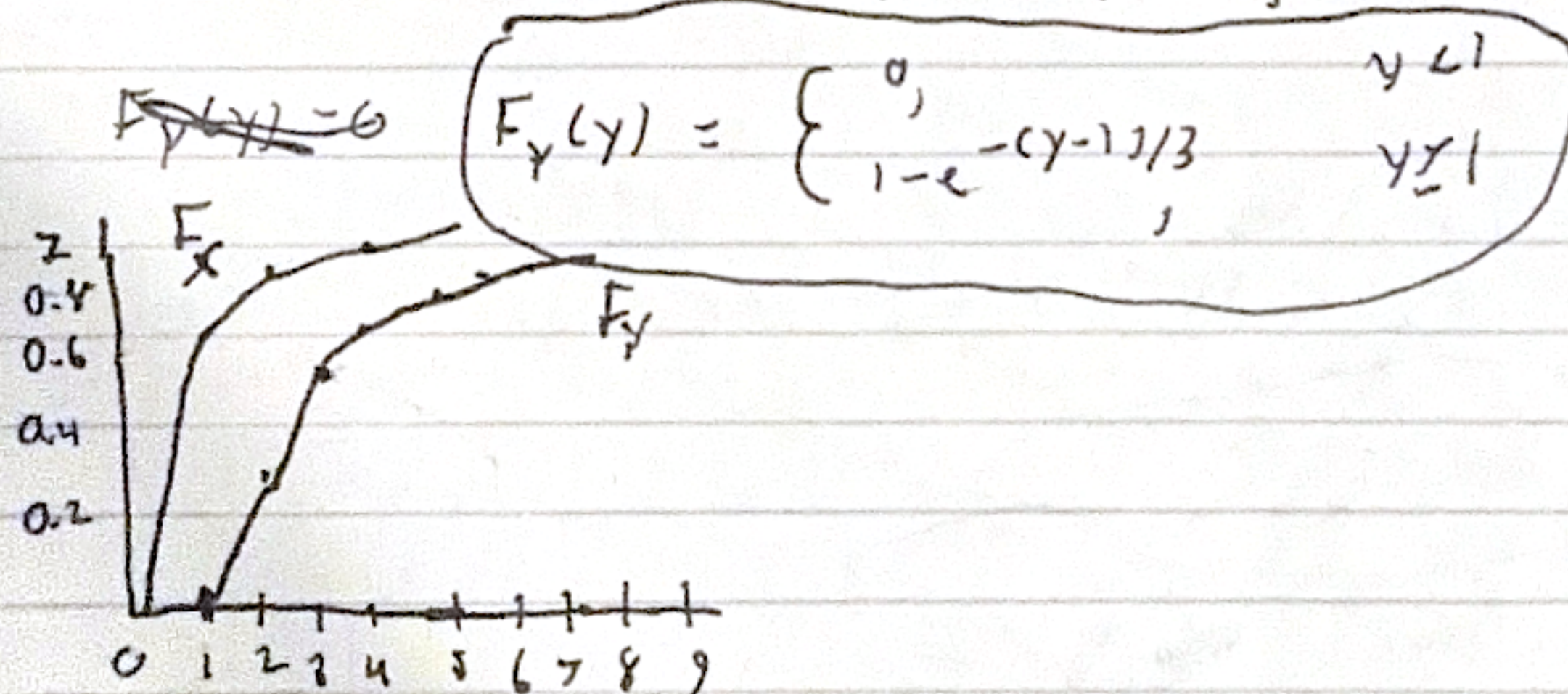
$$\frac{1}{e^{3.5}} \approx 0$$

~~37)~~ $P(X \leq 3.5) = 1 - e^{-3.5} = 1 - \frac{1}{e^{3.5}} \approx 1$

38) $Y = 1 + 3X$

$$F_Y(y) = P(Y \leq y) = P(1 + 3X \leq y) \quad 3X \leq y - 1 \quad X \leq \frac{y-1}{3}$$

$$F_Y(y) = P(X \leq \frac{y-1}{3}) \quad F_Y(y) = F_X\left(\frac{y-1}{3}\right)$$



shifted 1 unit to the right and 3 times flatter

$$P(Y < 10)$$

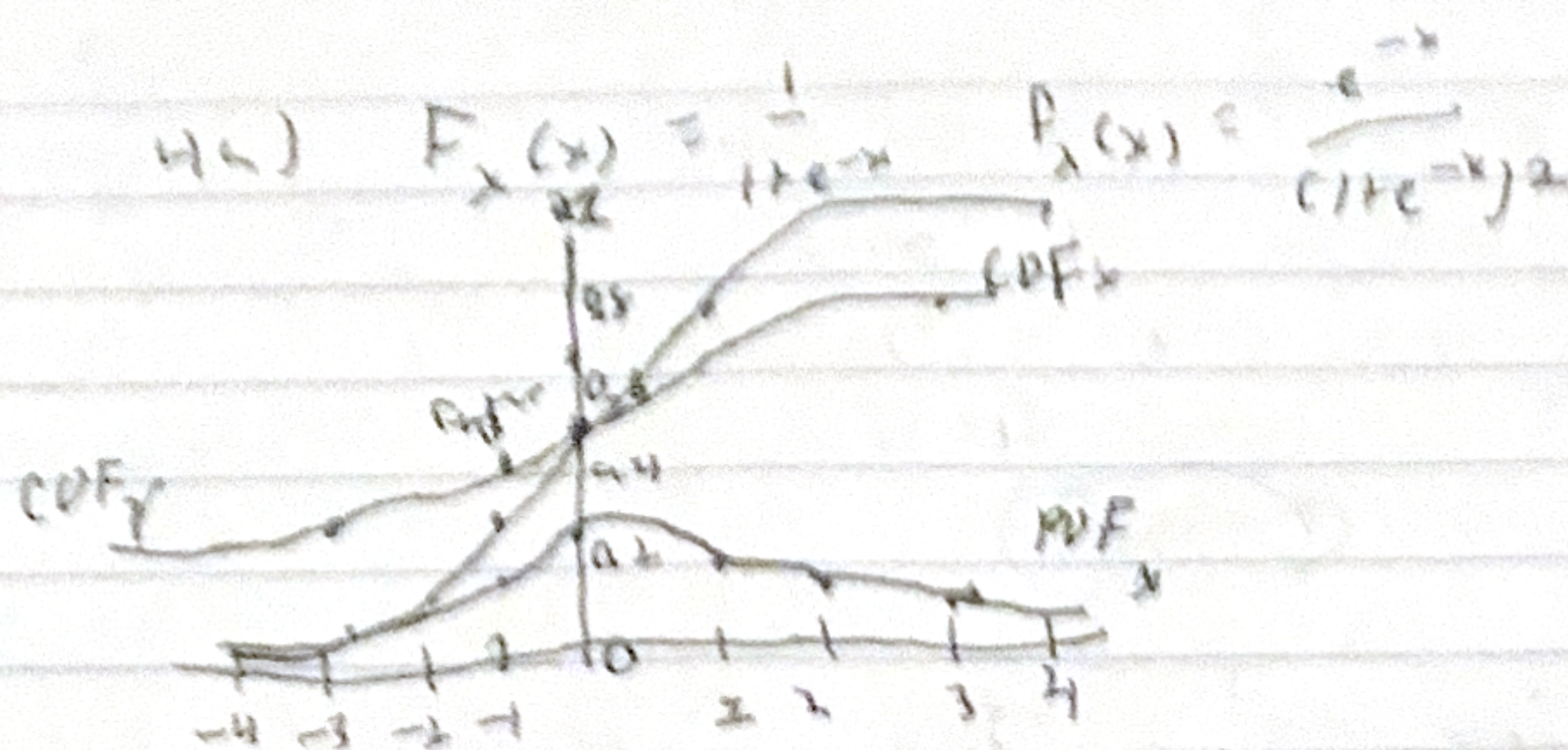
$$P(1 + 3X < 10)$$

$$3X < 9 \quad X < 3$$

$$P(X < 3) = 1 - e^{-3}$$

$$P(Y < 10) = 1 - e^{-3}$$

$$1 - \frac{1}{e^3}$$



46) $P(X \geq 0.8) = 1 - F_X(0.8) = 1 - \frac{1}{1+e^{-0.8}} = 1 - \frac{1}{1+\frac{1}{e^{0.8}}} = 1 - \frac{e^{0.8}}{e^{0.8}+1} = \frac{1}{e^{0.8}+1}$

$P(X \geq 0.8) = \frac{1}{e^{0.8}+1}$

$P(X \geq 3) = F_X(0.3) = \frac{1}{1+e^{-3}} = \frac{e^3}{e^3+1}$

$\frac{e^{0.8}+1}{e^{0.8}+1} - \frac{e^3}{e^3+1} = \frac{1}{e^{0.8}+1}$

46) $Y = 2X - 1$ $F_Y(y) = P(2X - 1 \leq y)$ $2X \leq y + 1$
 $= P(X \leq \frac{y+1}{2})$ $X \leq \frac{y+1}{2}$

$F_Y(y) = \frac{1}{1+e^{-(y+1)/2}}$

The new CDF is more centered at -1 and much flatter.

$P(Y < 0) = P(2X - 1 < 0)$ $X < 1/2$

$P(Y < 0) = \frac{1}{1+e^{-0.5}}$

$$54) F_x(x) = 1 - e^{-x} \quad 1 - e^{-m} = 0.5$$

$$-e^{-m} = 0.5$$

$$e^{-m} = 0.5$$

$$-m = \ln(0.5)$$

$$m = \ln(2)$$

$$56) F_x(x) = \frac{1}{1+e^{-x}} \quad \frac{1}{1+e^{-m}} = 0.5$$

$$1 = 0.5(1+e^{-m})$$

$$2 = 1+e^{-m}$$

$$1 = e^{-m} \quad m = 0$$

$$57) p = 1 - e^{-x}$$

$$1-p = e^{-x}$$

$$\ln(1-p) = -x$$

OLP21

$$Q(p) = -\ln(1-p)$$

$$Q(0.5) = -\ln(1-0.5) = -\ln(0.5) = \ln(2)$$

$$58) p = \frac{1}{1+e^{-x}}$$

$$p(1+e^{-x}) = 1$$

$$1+e^{-x} = \frac{1}{p}$$

OLP21

$$e^{-x} = \frac{1}{p} - 1 = \frac{1-p}{p}$$

$$Q(p) = \ln\left(\frac{1}{1-p}\right)$$

$$-x = \ln\left(\frac{1-p}{p}\right)$$

```
In [6]: import pandas as pd
metabric = pd.read_csv("metabric.csv")
metabric.head()
```

Out[6]:

	Age at Diagnosis	Type of Breast Surgery	Cancer Type	Chemotherapy	Hormone Therapy	Lymph nodes examined positive	Mutation Count
0	43.19	BREAST CONSERVING	Breast Cancer	NO	YES	0.0	2.0
1	48.87	MASTECTOMY	Breast Cancer	YES	YES	1.0	2.0
2	47.68	MASTECTOMY	Breast Cancer	YES	YES	3.0	1.0
3	76.97	MASTECTOMY	Breast Cancer	YES	YES	8.0	2.0
4	78.77	MASTECTOMY	Breast Cancer	NO	YES	0.0	4.0

In []:

```
In [9]: #5A and 5B
import numpy as np
import matplotlib.pyplot as plt

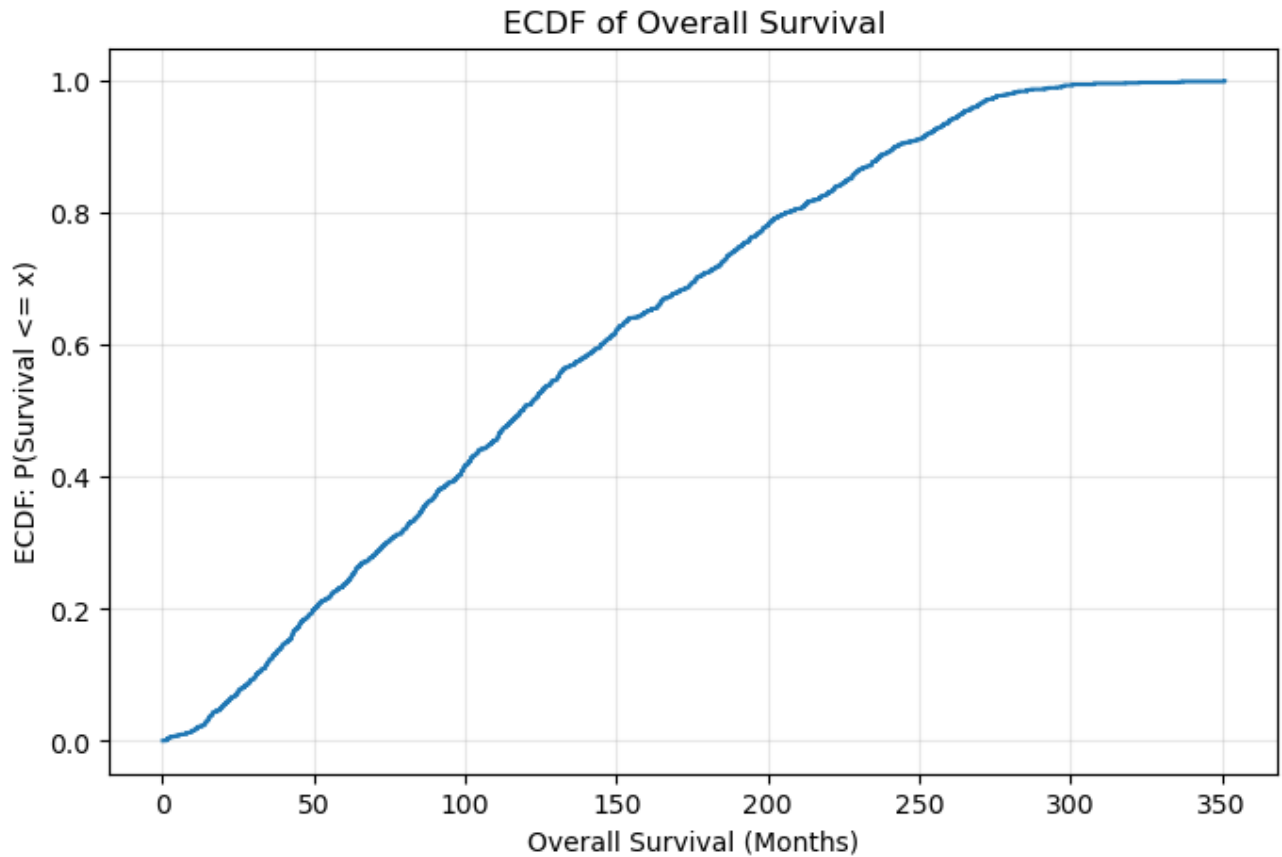
survival = metabric[["Overall Survival (Months)", "Chemotherapy"]].dropna()

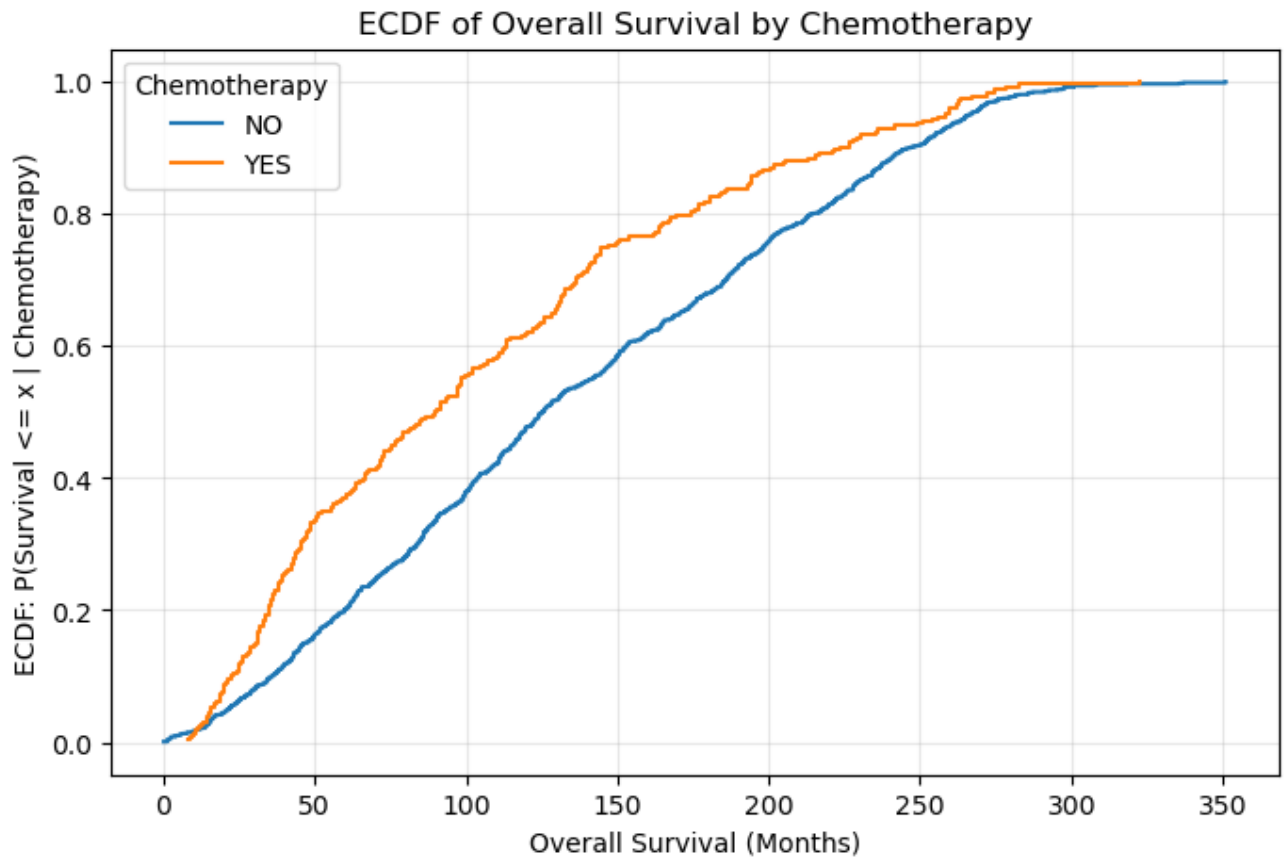
survival_times = np.sort(survival["Overall Survival (Months)"].to_numpy())
probabilities = np.arange(1, len(survival_times) + 1) / len(survival_times)

plt.figure(figsize=(8, 5))
plt.step(survival_times, probabilities, where="post")
plt.xlabel("Overall Survival (Months)")
plt.ylabel("ECDF: P(Survival <= x)")
plt.title("ECDF of Overall Survival")
plt.grid(alpha=0.3)
plt.show()

plt.figure(figsize=(8, 5))
for chemotherapy_status, group in survival.groupby("Chemotherapy", sort=True):
    group_times = np.sort(group["Overall Survival (Months)"].to_numpy())
    group_probabilities = np.arange(1, len(group_times) + 1) / len(group_times)
```

```
plt.step(group_times, group_probabilities, where="post", label=chemother  
  
plt.xlabel("Overall Survival (Months)")  
plt.ylabel("ECDF: P(Survival <= x | Chemotherapy)")  
plt.title("ECDF of Overall Survival by Chemotherapy")  
plt.legend(title="Chemotherapy")  
plt.grid(alpha=0.3)  
plt.show()
```





5B

Shorter amount of time, since the chemo line graph is to the left of the non chemo group for every x value, that means that for every value of months msubi there will be a greater or equal proportion of people deceased in the chemo group than the non chemo group.

5C

The data does not tell us this question. From the graph it looks like a harmful treatment but this is because our data from each group does not come from the same population. The people in the chemo group most likely have higher stages of cancer than the people in the non chemo group. Their doctors recommended that they take chemo because their cancer status was probably very life threatening. Since our samples are not from the same population we can not conduct a causal analysis. We would need to randomly assign people in to chemo and non chemo to create a causal inference.